

UNIT - 1

a) Theory :

- i) History of Java
- ii) POP vs OOP
- iii) Features, Advantages and Disadvantages
- iv) JDK, JVM and JRE
- v) Program structure and Execution Flow

b) Basic Understanding :

- i) Keywords, Variables, Datatypes
- ii) Wrapper classes - Boxing and Unboxing, Autoboxing and Autounboxing
- iii) Type Casting
- iv) Operators and Packages
- v) Methods
- vi) Java Input-Output

c) Statements :

- i) Conditional - if, if-else, switch
- ii) Looping - for, while, do-while
- iii) Transfer - break, continue, goto, return

d) String :

- i) Ways or creating strings
- ii) String methods

e) Arrays :

- i) Declaration, creation and initialization
- ii) Types : Single and Multi dimensional
- iii) Extra :
 - 1) Anonymous arrays
 - 2) Cloning of arrays
 - 3) Jagged Arrays
 - 4) Adv and Disadv of Arrays
 - 5) Array class - Helper class

UNIT - 2

a) Introduction :

- i) OOP Concepts
- ii) **Classes, Objects and Methods in a Java Class**
- iii) Ways to initialize an object in Java

b) Constructors :

- i) Introduction
- ii) **Types**

c) Inheritance :

- i) Introduction
- ii) **Types :**
 - 1) **Single - How JVM Works**
 - 2) **Multi-Level - Super Keyword**
 - 3) **Multiple - Diamond Problem**
 - (a) **Interfaces**
 - (b) **Implements vs Extends**
 - 4) **Hierarchical**
 - 5) **Hybrid**
- iii) **Abstract Classes :**
 - 1) **Introduction**
 - 2) **How ?**
 - 3) **Advantages**

d) Polymorphism :

- i) Introduction
- ii) **Types :**
 - 1) **Compile Time**
 - 2) **Run Time**

e) Exception Handling :

- i) Introduction
- ii) **Types : Checked, Unchecked, Error and Exception**
- iii) **Blocks :**
 - 1) **Try**
 - 2) **Catch**
 - 3) **Nested**
 - 4) **Finally**
 - 5) **Throw**
 - 6) **Throws**
- iv) **Types of exceptions :**
 - 1) **Arithmetic**
 - 2) **NullPointerException**
 - 3) **NumberFormatException**
 - 4) **ArrayIndexOutOfBoundsException**
 - 5) **StringIndexOutOfBoundsException**

- f) Threads :
 - i) Introduction and Types
 - ii) How ?
 - iii) Methods
 - iv) State transition Diagram
 - v) Thread vs Runnable

UNIT - 3

- a) Annotations : Built-in Annotations, Meta Annotations, Custom Annotations

- b) JDBC Drivers (any 1 or all) :

- i) Type 1
- ii) Type 2
- iii) Type 3
- iv) Type 4

- c) JDBC Connectivity : Steps

- d) SQL Basics :

- i) Types :
 - DDL (Structure) : CREATE, ALTER, DROP TRUNCATE
 - DML (Data) : SELECT, INSERT, UPDATE, DELETE
 - DCL : GRANT, REVOKE
 - TCL : COMMIT, ROLLBACK, SAVEPOINT
- CRUD Operations :
 - Create, Read, Update, Delete - Examples

- e) Theory :

- i) JDBC Architecture : Flow
- ii) Important JDBC Classes
- iii) Importance of JDBC

- f) Programs :

- JDBC Connection
- Insertion
- Selection
- Updation
- Deletion

UNIT - 4

a) Introduction :

- i) GUI Programming : Definition and Examples
- ii) AWT vs Swing - Differences
- iii) Swing : What ? Features
- iv) Swing Class Hierarchy

b) GUI Components :

- JFrame
- JLabel
- JTextField
- JTextArea
- JButton
- JCheckBox
- JRadioButton
- JComboBox
- JList

c) Swing Containers

- Types :
 - JFrame
 - JDialog
 - JPanel

d) JFrame :

- Syntax
- Important Methods :
 - setSize()
 - setLayout()
 - setVisible()
 - setDefaultCloseOperation()
 - add()

e) Event Handling :

- Mouse Events
- Keyboard Events